

EASY BUILD: AI AND RAG TRANSFORMATION STRATEGY

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1. Enterprise Search & RAG Roadmap

This strategic document outlines our deployment of Retrieval-Augmented Generation (RAG) models to improve searchability of internal catalogs, employee handbooks, and operations manuals. Our target is to build a unified interface using Qwen2-VL for layout-aware parsing and Ollama for embeddings.

Data pipelines ingest real-time sales transactions, process them via stream processors, and sync them to PostgreSQL and ChromaDB databases for reporting and analytics.

Specifically, the microservices architecture employs Kong API Gateway for routing client calls, securing endpoints via OAuth 2.0, and applying rate limits to protect backend services from denial of service.

Consequently, the Order Management service interacts with the ERP core database over port 5439. The database runs on PostgreSQL 15.4 to handle complex ACID transactions and structured JSON metadata queries.

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2. Multimodal OCR and Vision Indexing

RAG pipelines often fail when critical data is stored inside drawings, charts, or diagrams. To solve this, our system routes layout chunks with visual components to a local vision-language model, which generates detailed text summaries to enrich the vector store indices.

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3. Vector Databases and Rerankers

We utilize ChromaDB with cosine similarity metrics for dense vector retrieval. To handle complex queries, we employ hybrid search (ChromaDB + BM25 keyword matching) combined with cross-encoder rerankers like BAAI/bge-reranker-base.

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